

ICU Mortality Prediction

Modeling Phase

Preprocessing, model development, comparison,
and final test evaluation

0.852

Final test ROC-AUC

0.682

Final test recall

0.600

Classification threshold

Final candidate

- Deep Learning was selected for recall priority and retained recall-focused behavior on the test set.

1. Preprocessing strategy

Prepare clean, consistent inputs for model development.

- The final dataset contains 45,253 ICU stays.
- Use 83 predictors: 80 numeric and 3 categorical features.
- Categorical variables: gender, admission_type, admission_location.
- Fill numeric missing values with median imputation.
- Encode categorical variables with OneHotEncoder.
- Use StandardScaler only for scale-sensitive models: Logistic Regression, KNN, and Deep Learning.

3

Categorical features

80

Numeric features

36,169 x 83

Training matrix

2. Train/test split and leakage control

Use patient-level grouping for model evaluation.

- Create an approximately 80/20 training/test split.
- Split at SUBJECT_ID level.
- Keep different ICU stays from the same patient on the same side of the split.
- The prediction unit is ICU stay; split control is patient-level.
- Exclude SUBJECT_ID and ICUSTAY_ID from predictive features.
- Reserve the test set for final evaluation.

Key idea

- Placing the same patient in both sets can expose the model to that patient during training.
- Patient-level grouping is therefore a central leakage control.

3. Modeling and evaluation plan

Compare models with a shared set of performance metrics.

Models evaluated

- Logistic Regression
- KNN
- Decision Tree
- Random Forest
- XGBoost
- Deep Learning

Improvement methods

- Hyperparameter tuning
- Class weights / `scale_pos_weight`
- SMOTE
- Threshold tuning
- EarlyStopping

Metrics

- ROC-AUC: discrimination
- Precision: fraction of positive predictions that are correct
- Recall: fraction of mortality cases detected
- F1: precision-recall balance

4. Logistic Regression

Baseline model and threshold tuning

Stage	ROC-AUC	Precision	Recall	F1
Baseline	0.842	0.644	0.239	0.349
Balanced	0.843	0.311	0.741	0.438
Threshold 0.2	0.842	0.400	0.570	0.470

- The baseline had good ROC-AUC but low recall.
- Class weighting increased recall at a substantial precision cost.
- Threshold 0.2 on standard Logistic Regression gave a more suitable recall-focused trade-off.
- Retain this model as the reference baseline.

0.400

Selected precision

0.570

Selected recall

0.470

Selected F1

5. KNN

Class imbalance in a distance-based model

Stage	ROC-AUC	Precision	Recall	F1
Baseline K=5	0.709	0.614	0.129	0.213
K=3	0.667	0.523	0.167	0.253
SMOTE + K=3	0.739	0.236	0.708	0.353

- StandardScaler is used because KNN relies on distances.
- Different K values were tested; recall declined as K increased.
- SMOTE strongly increased recall but sharply reduced precision.
- KNN was less balanced than the strongest final candidates.

0.236

SMOTE precision

0.708

SMOTE recall

0.353

SMOTE F1

6. Decision Tree

Hyperparameter tuning and class balancing

Stage	ROC-AUC	Precision	Recall	F1
Baseline	0.621	0.323	0.339	0.331
Tuned	0.760	0.577	0.142	0.226
Tuned + balanced	0.772	0.253	0.701	0.371
Threshold 0.6	0.772	0.312	0.536	0.394

- GridSearchCV controlled excessive tree growth.
- Class weighting raised recall, while precision remained limited.
- Threshold 0.6 provided a more acceptable precision-recall balance.

Best parameters

- `max_depth = 5`
- `min_samples_leaf = 5`
- `min_samples_split = 2`

7. Random Forest

More stable results with a tree ensemble

Stage	ROC-AUC	Precision	Recall	F1
Baseline	0.846	0.741	0.159	0.262
Tuned	0.855	0.798	0.135	0.231
Tuned + balanced	0.862	0.455	0.552	0.498

- The baseline had good ROC-AUC and precision, but low recall.
- Tuning increased precision but reduced recall further.
- Balanced class weighting meaningfully improved recall and F1.

Best parameters

- `n_estimators = 200`
- `max_depth = None`
- `min_samples_leaf = 5`
- `max_features = sqrt`

8. XGBoost

A strong precision-recall balance with boosting

Stage	ROC-AUC	Precision	Recall	F1
Baseline	0.855	0.606	0.314	0.413
Tuned	0.870	0.666	0.300	0.414
scale_pos_weight	0.864	0.382	0.680	0.489
Threshold 0.6	0.864	0.446	0.587	0.507

- Hyperparameter tuning improved ROC-AUC.
- scale_pos_weight gave greater weight to mortality-positive cases.
- Threshold 0.6 produced the strongest overall balance and highest F1 among the final candidates.

Best parameters

- n_estimators = 200
- max_depth = 5
- learning_rate = 0.1
- scale_pos_weight ~ 7.36

9. Deep Learning

The final candidate best aligned with recall priority

Stage	ROC-AUC	Precision	Recall	F1
Baseline	0.864	0.624	0.325	0.427
Class weighting	0.868	0.350	0.753	0.477
Threshold 0.6	0.868	0.404	0.651	0.499

- Use a small feed-forward neural network for tabular data.
- Class weighting strongly improved recall.
- Threshold 0.6 was selected for the recall-focused objective.

Architecture & training

- 64 -> 32 -> 1
- ReLU hidden layers
- Sigmoid output
- Adam + binary cross-entropy
- Batch size 64
- EarlyStopping

10. All models compared

Precision, recall, and F1 for the selected version of each model.

Model comparison: precision, recall, and F1



11. Final model candidates

Random Forest, XGBoost, and Deep Learning stood out.

Final candidates



Model	AUC	Prec.	Recall	F1
Random Forest	0.862	0.455	0.552	0.498
XGBoost	0.864	0.446	0.587	0.507
Deep Learning	0.868	0.404	0.651	0.499

- Random Forest had the highest precision among the final candidates.
- XGBoost achieved the highest F1.
- Deep Learning had the highest recall among these candidates and best matched the clinical objective.

12. Final test evaluation

Evaluate the selected Deep Learning model on the untouched test set.

0.852

ROC-AUC

0.347

Precision

0.682

Recall

0.460

F1

- Train the final model on the full training set using class weights.
- Apply the validation-selected threshold of 0.6 without changing it on the test set.
- Use the test set only at the final evaluation stage.
- No dramatic performance drop was observed from validation to test.
- The model retained its recall-focused behavior.

Final decision

- Model: Deep Learning
- Rationale: prioritize detection of mortality cases
- Final test recall: 0.682